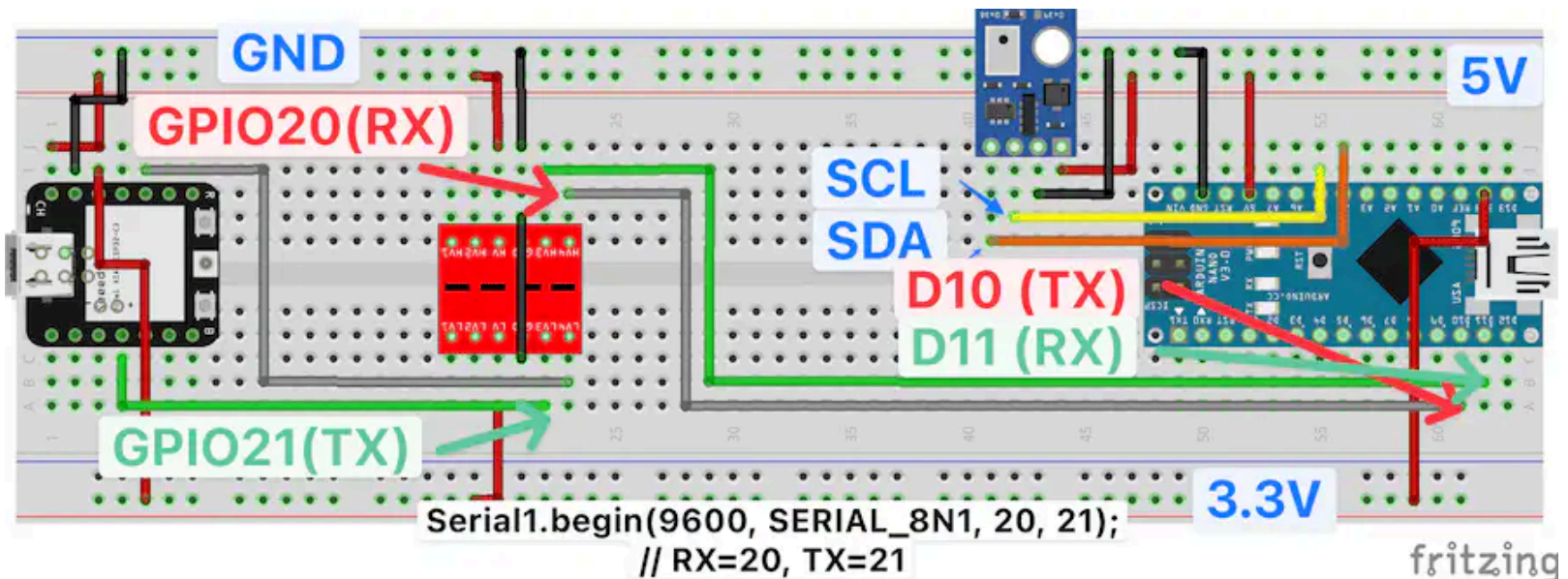


The Story of CSC 494

IoT Software/Hardware of CSC 494



- IoT Project Skeleton will be provided.
 - Ardunio Nano for controller
 - ESP32C6 for BLE/WiFi
 - TMP102 I2C Temperature Sensor

- Hardware: We will discuss basic concepts/tools of IoT (Embedded Systems) as needed.
- Software: We will discuss software stacks for acquiring, processing, and sharing sensor information with any computer systems.

The Obligation of Using Provided Hardware

- To quickly get started, I provide students with the project skeleton hardware and software.
- Students can purchase the same hardware from anywhere without using the provided hardware.
- In the case students use the provided hardware, they are obligated to return the hardware at the end of the course.

- Students must take good care of the provided hardware during the course and return everything in good and working condition.
- Students who lose or damage the provided hardware must purchase the same hardware and return it to the instructor.
- Students who fail to return the hardware or fail to purchase the replacement hardware will receive F for this course.

- Students who drop this course must return the provided hardware.
- If students fail to return the hardware after dropping the course, they can be reported as theft of school property to the university authorities.

Purchase of Additional Hardware

- Students do not need to purchase any additional hardware components if they can implement their project ideas using the provided hardware.
- However, depending on the project ideas, students may need to purchase additional hardware components.
- Students are responsible for purchasing any additional hardware components needed for their projects.

Purchase of LLM Tools

- Students may need to purchase access to LLM tools (e.g., ChatGPT, Claude, Gemini, etc.) for their projects.
- Students are responsible for purchasing any LLM tools needed for their projects.

Evaluation

1. IoT Project (40%)

- You will evaluate three peers: artifacts and progress
- You will be evaluated by three peers: artifacts and progress

2. Learning two IoT topics with AI (20%)

- You will evaluate three peers: read their documents and process
- You will share your learning documents and process, and evaluated by three peers

3. One examination (25%)

- On Week 14 Wednesday (online)

4. Three Assignments (HW1 - HW3) (15%)

HW1 - HW3: Canvas submission

Problem Solving Project Timeline

Three Stages:

1. Stage 1: Preparation
2. Stage 2: Prototype/MVP
3. Stage 3: Project

- This is a guideline and ceremonial: you can adjust the timeline as needed.
- Start early to finish early if you already started the project.

Stage 1: Preparation (Weeks 1-3)

- I'll Make a Canvas Page for your project (by the end of week 1)
- I'll let you know whom you are going to evaluate using email(by the end of week 1)
 - You **should not know** the peers who evaluate you.

- Define the problem to solve (by the end of week 2):
HW1
 - Make GitHub Repository for your project
 - Make README.md for your project
 - Upload your plans for your peers

- Make ready for the PPP: HW2
 - Write your PPP document and upload it to your GitHub Repository (by the end of week 3)
 - Read your peers' PPP documents (by the end of week 3) for evaluation

Stage 2: Prototype/MVP (Weeks 4-8)

- PPP (Project Planning Presentation) (First meeting of Week 4)
- 1 or 2 presentations for project, and the other 1 or 2 presentation for learning with AI*
- S1R (Sprint 1 Retrospective) & S1P (Sprint 1 Presentation) (Last meeting of Week 8)

**The same format as the individual project presentation (ASE format).*

Stage 3: Project (Weeks 9-16)

- Work on your project
- 1 - 3 presentations for project, and the other 1 - 3 presentations for learning with AI*
- S2R (Sprint 2 Retrospective) & FP (Final Presentation at the last week)

**The same format as the individual project presentation.*

Class Schedule

- Two meetings per week (Total 3 hours)
- First part of the class: presentations by students (IoT project and learning with AI)
- Second part of the class: discussions of important IoT topics

Tentative Schedule of Each Class Meeting

Stage 1

- Mainly project discussion

Stage 2

- IoT Topics to Discuss

Stage 3

- Focus mainly on project, possibly individual project discussion and such.

Possible IoT Topics to Discuss

Planned

1. Circuits
2. Microcontrollers and Embedded Systems
3. IoT Communication Protocols
4. Sensors & Actuators

Many more topics can be added as needed.

Possible AI/LLM Topics to Discuss

1. The 2nd Brain System - Tools & Rules
2. Understanding how AI and LLM work
3. Using AI/LLMs for Problem Solving

Many more topics can be added as needed.